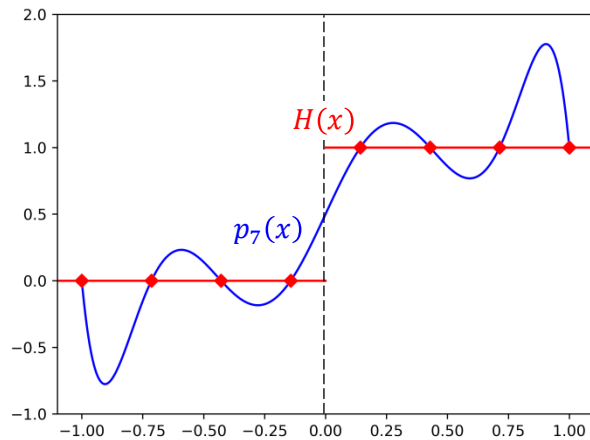
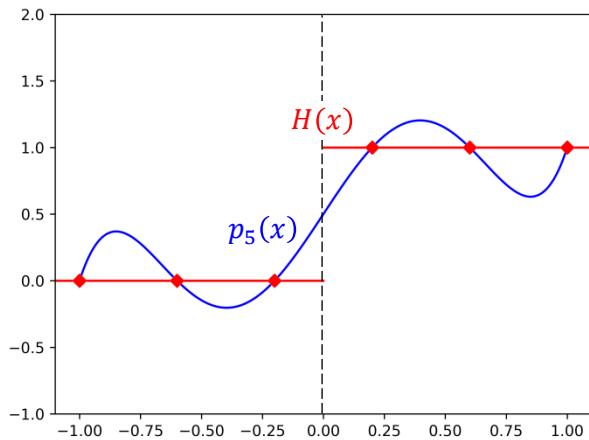
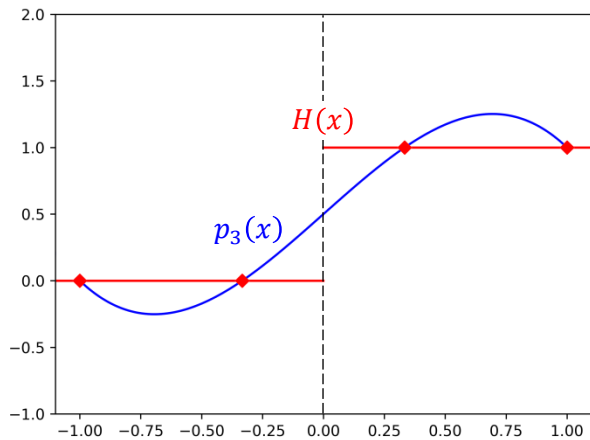


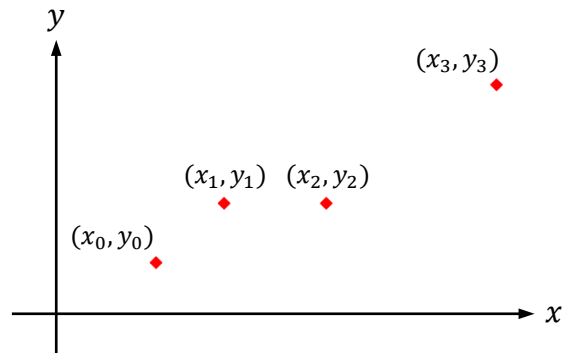
Spline Interpolation

Heaviside Step Function

$$H(x) = \begin{cases} 0, & x < 0 \\ 1, & x > 0 \end{cases}$$



Spline Interpolation



Linear Spline Interpolation

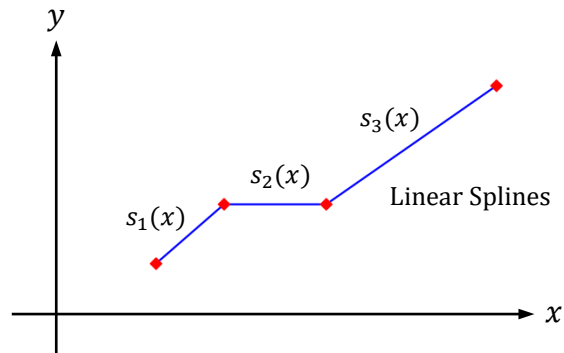
$$s_1(x) = a_1x + b_1; \quad x_0 \leq x \leq x_1$$

$$s_2(x) = a_2x + b_2; \quad x_1 \leq x \leq x_2$$

$$s_3(x) = a_3x + b_3; \quad x_2 \leq x \leq x_3$$

$$\vdots$$

$$s_i(x) = a_ix + b_i; \quad x_{i-1} \leq x \leq x_i, i = 1, 2, \dots, n$$

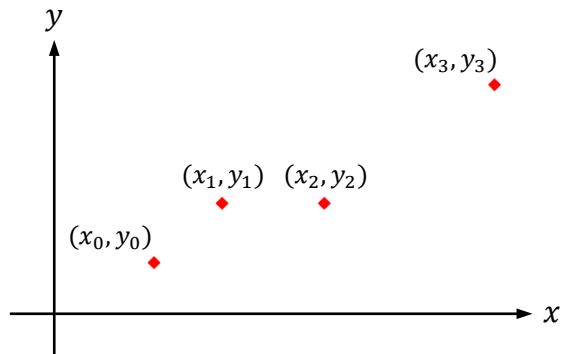


$$\left. \begin{array}{l} s_i(x_{i-1}) = f(x_{i-1}) \rightarrow f(x_{i-1}) = a_i \cdot x_{i-1} + b_i \\ s_i(x_i) = f(x_i) \rightarrow f(x_i) = a_i \cdot x_i + b_i \end{array} \right\} i = 1, 2, \dots, n$$

$$a_i = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}}$$

$$\begin{aligned} b_i &= f(x_{i-1}) - \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \cdot x_{i-1} \\ &= f(x_i) - \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \cdot x_i \end{aligned}$$

Spline Interpolation



Quadratic Spline Interpolation

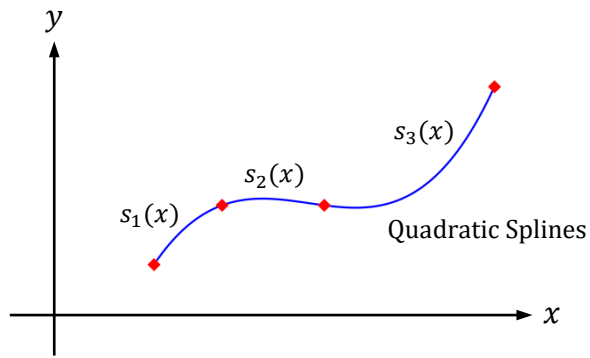
$$s_1(x) = a_1x^2 + b_1x + c_1; \quad x_0 \leq x \leq x_1$$

$$s_2(x) = a_2x^2 + b_2x + c_2; \quad x_1 \leq x \leq x_2$$

$$s_3(x) = a_3x^2 + b_3x + c_3; \quad x_2 \leq x \leq x_3$$

$$\vdots$$

$$s_i(x) = a_ix^2 + b_ix + c_i; \quad x_{i-1} \leq x \leq x_i, i = 1, 2, \dots, n$$



$$s_1(x_0) = f(x_0) \quad \rightarrow \quad f(x_0) = a_1 \cdot x_0^2 + b_1 \cdot x_0 + c_1$$

$$s_i(x_i) = f(x_i) \quad \rightarrow \quad f(x_i) = a_i \cdot x_i^2 + b_i \cdot x_i + c_i$$

$$s'_i(x_i) = s'_{i+1}(x_i) \quad \rightarrow \quad 2a_i \cdot x_i + b_i = 2a_{i+1} \cdot x_i + b_{i+1}$$

$$s_{i+1}(x_i) = f(x_i) \quad \rightarrow \quad f(x_i) = a_{i+1} \cdot x_i^2 + b_{i+1} \cdot x_i + c_{i+1}$$

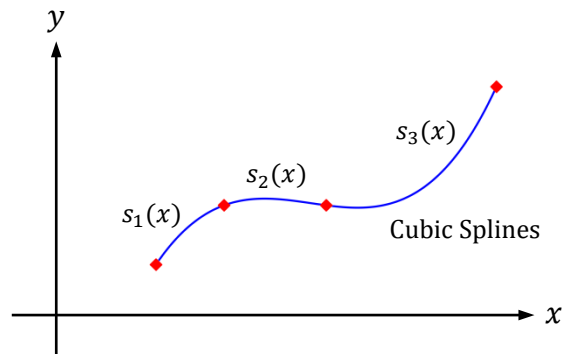
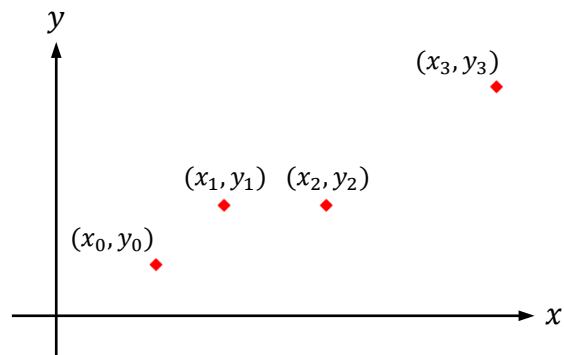
$$s_n(x_n) = f(x_n) \quad \rightarrow \quad f(x_n) = a_n \cdot x_n^2 + b_n \cdot x_n + c_n$$

$$i = 1, 2, \dots, n-1$$

$3n - 1$ equations for $3n$ unknowns (a_i, b_i, c_i for $i = 1, 2, \dots, n$).

Which one should be specified ???

Spline Interpolation



Cubic Spline Interpolation

$$s_1(x) = a_1x^3 + b_1x^2 + c_1x + d_1; \quad x_0 \leq x \leq x_1$$

$$s_2(x) = a_2x^3 + b_2x^2 + c_2x + d_2; \quad x_1 \leq x \leq x_2$$

$$s_3(x) = a_3x^3 + b_3x^2 + c_3x + d_3; \quad x_2 \leq x \leq x_3$$

$\vdots$

$$s_i(x) = a_ix^3 + b_ix^2 + c_ix + d_i; \quad x_{i-1} \leq x \leq x_i, i = 1, 2, \dots, n$$

$$s_0''(x_0) = 0 \quad \rightarrow$$

$$6a_1 \cdot x_0 + 2b_1 = 0$$

$$s_1(x_0) = f(x_0) \quad \rightarrow$$

$$f(x_0) = a_1 \cdot x_0^3 + b_1 \cdot x_0^2 + c_1 \cdot x_0 + d_1$$

$$s_i(x_i) = f(x_i) \quad \rightarrow$$

$$f(x_i) = a_i \cdot x_i^3 + b_i \cdot x_i^2 + c_i \cdot x_i + d_i$$

$$s_i'(x_i) = s_{i+1}'(x_i) \quad \rightarrow$$

$$3a_i \cdot x_i^2 + 2b_i \cdot x_i + c_i = 6a_{i+1} \cdot x_i^2 + 2b_{i+1} \cdot x_i + c_{i+1}$$

$$s_i''(x_i) = s_{i+1}''(x_i) \quad \rightarrow$$

$$6a_i \cdot x_i + 2b_i = 6a_{i+1} \cdot x_i + 2b_{i+1}$$

$$s_{i+1}(x_i) = f(x_i) \quad \rightarrow$$

$$f(x_i) = a_{i+1} \cdot x_i^3 + b_{i+1} \cdot x_i^2 + c_{i+1} \cdot x_i + d_{i+1}$$

$$s_n(x_n) = f(x_n) \quad \rightarrow$$

$$f(x_n) = a_n \cdot x_n^3 + b_n \cdot x_n^2 + c_n \cdot x_n + d_n$$

$$s_n''(x_n) = 0 \quad \rightarrow$$

$$6a_n \cdot x_n + 2b_n = 0$$

$i = 1, 2, \dots, n-1$

$4n$ equations for $4n$ unknowns (a_i, b_i, c_i, d_i for $i = 1, 2, \dots, n$).

This leads to large system of equations.

Spline Interpolation

Modified Cubic Spline Interpolation

Lagrange Polynomials:

$$s_i''(x) = \frac{x - x_i}{x_{i-1} - x_i} \cdot f''(x_{i-1}) + \frac{x - x_{i-1}}{x_i - x_{i-1}} \cdot f''(x_i); \quad x_{i-1} \leq x \leq x_i$$

Integration Twice:

$$s_i(x) = \frac{1}{6} \cdot \frac{(x - x_i)^3}{x_{i-1} - x_i} \cdot f''(x_{i-1}) + \frac{1}{6} \cdot \frac{(x - x_{i-1})^3}{x_i - x_{i-1}} \cdot f''(x_i) + C_1 \cdot x + C_2; \quad x_{i-1} \leq x \leq x_i$$

$$s_i(x_{i-1}) = f(x_{i-1})$$

$$s_i(x_i) = f(x_i)$$

$$C_1 = \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} - \frac{1}{6} \cdot [f''(x_i) - f''(x_{i-1})] \cdot (x_i - x_{i-1})$$

$$C_2 = \frac{x_i}{x_i - x_{i-1}} \cdot f(x_{i-1}) + \frac{x_{i-1}}{x_{i-1} - x_i} \cdot f(x_i) + \frac{x_i}{6} \cdot (x_{i-1} - x_i) \cdot f''(x_{i-1}) + \frac{x_{i-1}}{6} \cdot (x_i - x_{i-1}) \cdot f''(x_i)$$

$$s_i(x) = \frac{1}{6} \cdot \frac{(x - x_i)^3}{x_{i-1} - x_i} \cdot f''(x_{i-1}) + \frac{1}{6} \cdot \frac{(x - x_{i-1})^3}{x_i - x_{i-1}} \cdot f''(x_i)$$

$$+ \left[\frac{f(x_i)}{x_i - x_{i-1}} - \frac{f''(x_i)}{6} \cdot (x_i - x_{i-1}) \right] \cdot (x - x_{i-1}) + \left[\frac{f(x_{i-1})}{x_i - x_{i-1}} - \frac{f''(x_{i-1})}{6} \cdot (x_i - x_{i-1}) \right] \cdot (x_i - x); \quad x_{i-1} \leq x \leq x_i$$

Spline Interpolation

Modified Cubic Spline Interpolation

Compatibility:

$$s'_i(x_i) = s'_{i+1}(x_i)$$

$$\begin{aligned} & \frac{1}{2} \cdot (x_i - x_{i-1}) \cdot f''(x_i) + \left[\frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} - \frac{f''(x_i) - f''(x_{i-1})}{6} \cdot (x_i - x_{i-1}) \right] \\ & \qquad \qquad \qquad (x_{i-1} \leq x \leq x_i) \\ & = \frac{1}{2} \cdot (x_i - x_{i+1}) \cdot f''(x_i) + \left[\frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} - \frac{f''(x_{i+1}) - f''(x_i)}{6} \cdot (x_{i+1} - x_i) \right] \\ & \qquad \qquad \qquad (x_i \leq x \leq x_{i+1}) \end{aligned}$$

Rearranging:

$$i = 1, 2, \dots, n-1$$

$$\begin{aligned} & (x_i - x_{i-1}) \cdot f''(x_{i-1}) + 2 \cdot (x_{i+1} - x_{i-1}) \cdot f''(x_i) + (x_{i+1} - x_i) \cdot f''(x_{i+1}) \\ & \qquad \qquad \qquad = 6 \cdot \left[\frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} - \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \right] \end{aligned}$$

System of Equations:

$$\begin{bmatrix} \times & & & & \\ \times & \times & \times & & \\ & \ddots & \ddots & \ddots & \\ & & \times & \times & \times \\ & & & & \times \end{bmatrix} \begin{bmatrix} f''(x_0) \\ f''(x_1) \\ \vdots \\ f''(x_{n-1}) \\ f''(x_n) \end{bmatrix} = \begin{bmatrix} \times \\ \times \\ \vdots \\ \times \\ \times \end{bmatrix}$$

Spline Interpolation

Example 04-03: Find modified cubic spline interpolation functions from the following data.

$f''(x_0) = 0$
 $f''(0.00) = 0$

$f''(x_n) = 0$
 $f''(0.60) = 0$

i	x_i	$f(x_i)$
0	0.00	0.000000
1	0.25	0.210368
2	0.40	0.399829
3	0.60	0.405278

$i = 1$

$$(0.25 - 0.00) \cdot f''(0.00) + 2 \cdot (0.40 - 0.00) \cdot f''(0.25) + (0.40 - 0.25) \cdot f''(0.40)$$
$$= 6 \cdot \left(\frac{0.399829 - 0.210368}{0.40 - 0.25} - \frac{0.210368 - 0.000000}{0.25 - 0.00} \right)$$
$$0.25 \cdot f''(0.00) + 0.80 \cdot f''(0.25) + 0.15 \cdot f''(0.40) = 2.529608$$

$i = 2$

$$(0.40 - 0.25) \cdot f''(0.25) + 2 \cdot (0.60 - 0.25) \cdot f''(0.40) + (0.60 - 0.40) \cdot f''(0.60)$$
$$= 6 \cdot \left(\frac{0.405278 - 0.399829}{0.60 - 0.40} - \frac{0.399829 - 0.210368}{0.40 - 0.25} \right)$$
$$0.15 \cdot f''(0.25) + 0.70 \cdot f''(0.40) + 0.20 \cdot f''(0.60) = -7.414970$$

$$(x_i - x_{i-1}) \cdot f''(x_{i-1}) + 2 \cdot (x_{i+1} - x_{i-1}) \cdot f''(x_{i-1}) + (x_{i+1} - x_i) \cdot f''(x_{i+1})$$
$$= 6 \cdot \left[\frac{f(x_{i+1}) - f(x_i)}{x_{i+1} - x_i} - \frac{f(x_i) - f(x_{i-1}))}{x_i - x_{i-1}} \right]$$

Spline Interpolation

Example 04-03: Find modified cubic spline interpolation functions from the following data.

$f''(x_0) = 0$
 $f''(0.00) = 0$

$f''(x_n) = 0$
 $f''(0.60) = 0$

i	x_i	$f(x_i)$
0	0.00	0.000000
1	0.25	0.210368
2	0.40	0.399829
3	0.60	0.405278

$i = 1$

$$(0.25 - 0.00) \cdot f''(0.00) + 2 \cdot (0.40 - 0.00) \cdot f''(0.25) + (0.40 - 0.25) \cdot f''(0.40)$$
$$= 6 \cdot \left(\frac{0.399829 - 0.210368}{0.40 - 0.25} - \frac{0.210368 - 0.000000}{0.25 - 0.00} \right)$$
$$0.25 \cdot f''(0.00) + 0.80 \cdot f''(0.25) + 0.15 \cdot f''(0.40) = 2.529608$$

$i = 2$

$$(0.40 - 0.25) \cdot f''(0.25) + 2 \cdot (0.60 - 0.25) \cdot f''(0.40) + (0.60 - 0.40) \cdot f''(0.60)$$
$$= 6 \cdot \left(\frac{0.405278 - 0.399829}{0.60 - 0.40} - \frac{0.399829 - 0.210368}{0.40 - 0.25} \right)$$
$$0.15 \cdot f''(0.25) + 0.70 \cdot f''(0.40) + 0.20 \cdot f''(0.60) = -7.414970$$

$$\begin{bmatrix} 1 & & & \\ 0.25 & 0.80 & 0.15 & \\ & 0.15 & 0.70 & 0.20 \\ & & & 1 \end{bmatrix} \begin{Bmatrix} f''(0.00) \\ f''(0.25) \\ f''(0.40) \\ f''(0.60) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 2.529608 \\ -7.414970 \\ 0 \end{Bmatrix}$$
$$f''(0.25) = 5.363667, f''(0.40) = -11.742172$$

Spline Interpolation

Example 04-03: Find modified cubic spline interpolation functions from the following data.

	i	x_i	$f(x_i)$
$f''(x_0) = 0$ $f''(0.00) = 0$	0	0.00	0.000000
	1	0.25	0.210368
	2	0.40	0.399829
$f''(x_n) = 0$ $f''(0.60) = 0$	3	0.60	0.405278

$$\begin{bmatrix} 1 & & & \\ 0.25 & 0.80 & 0.15 & \\ & 0.15 & 0.70 & 0.20 \\ & & & 1 \end{bmatrix} \begin{Bmatrix} f''(0.00) \\ f''(0.25) \\ f''(0.40) \\ f''(0.60) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 2.529608 \\ -7.414970 \\ 0 \end{Bmatrix}$$

$$f''(0.25) = 5.363667, f''(0.40) = -11.742172$$

$$0.00 \leq x \leq 0.25$$

$$s_1(x) = \frac{1}{6} \cdot \frac{(x - 0.00)^3}{0.25 - 0.00} \cdot (5.363667) + \left[\frac{0.210368}{0.25 - 0.00} - \frac{(5.363667)}{6} \cdot (0.25 - 0.00) \right] \cdot (x - 0.00)$$

$$0.25 \leq x \leq 0.40$$

$$s_2(x) = \frac{1}{6} \cdot \frac{(x - 0.00)^3}{0.25 - 0.40} \cdot (5.363667) + \frac{1}{6} \cdot \frac{(x - 0.25)^3}{0.40 - 0.25} \cdot (-11.742172) + \left[\frac{0.399829}{0.40 - 0.25} - \frac{(-11.742172)}{6} \cdot (0.40 - 0.25) \right] \cdot (x - 0.25) + \left[\frac{0.210368}{0.40 - 0.25} - \frac{(5.363667)}{6} \cdot (0.40 - 0.25) \right] \cdot (0.40 - x)$$

$$0.40 \leq x \leq 0.60$$

$$s_3(x) = \frac{1}{6} \cdot \frac{(x - 0.60)^3}{0.40 - 0.60} \cdot (-11.742172) + \frac{0.405278}{0.60 - 0.40} \cdot (x - 0.40) + \left[\frac{0.399829}{0.60 - 0.40} - \frac{(-11.742172)}{6} \cdot (0.60 - 0.40) \right] \cdot (0.60 - x)$$

Spline Interpolation

Example 04-03: Find modified cubic spline interpolation functions from the following data.

	i	x_i	$f(x_i)$
$f''(x_0) = 0$ $f''(0.00) = 0$	0	0.00	0.000000
	1	0.25	0.210368
	2	0.40	0.399829
$f''(x_n) = 0$ $f''(0.60) = 0$	3	0.60	0.405278

$$\begin{bmatrix} 1 & & & \\ 0.25 & 0.80 & 0.15 & \\ & 0.15 & 0.70 & 0.20 \\ & & 1 & \end{bmatrix} \begin{Bmatrix} f''(0.00) \\ f''(0.25) \\ f''(0.40) \\ f''(0.60) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 2.529608 \\ -7.414970 \\ 0 \end{Bmatrix}$$

$$f''(0.25) = 5.363667, f''(0.40) = -11.742172$$

